

1. Differential Shrine — Differential Equations
 - a. First-order differential equation general idea

Example 4

A rectangular tank has a base of 5 m^2 . Water is flowing into the tank at a constant rate of 10 litres per second ($= 0.01 \text{ m}^3 \text{ s}^{-1}$). At the same time, water is leaking out at a rate proportional to $\sqrt{h}$ where h is the depth of the water at time t . When the depth is 1 metre the water is leaking out at 2 litres per second. Find the differential equation describing the rate of change of h with time t .

If water comes in at rate $0.01 \text{ m}^3 \text{ s}^{-1}$ and the base area is 5 m^2 , the water coming in is increasing the depth at a rate of 0.002 m s^{-1} .

So the DE is $\frac{dh}{dt} = 0.002t - k\sqrt{h}$.

If water is leaking out at 2 litres per second ($0.002 \text{ m}^3 \text{ s}^{-1}$) then this is decreasing the depth at a rate of $\frac{0.002}{5} = 0.0004 \text{ m s}^{-1}$.

When $h = 1$, $k\sqrt{h} = 0.0004 \Rightarrow k = 0.0004$, giving $\frac{dh}{dt} = 0.002t - 0.0004\sqrt{h}$.

1. The rate at which body temperature, T , falls is proportional to the difference between the body temperature T and the temperature T_0 of the surroundings. Find a differential equation relating body temperature, T , and time t .
2. A certain substance is formed in a chemical reaction. The mass of the substance formed t seconds after the start of the reaction is x grams. At any time the rate of formation of the substance is proportional to $(30 - x)$. Find a differential equation relating x and t .
3. In studying the spread of a disease, a scientist thinks that the rate of infection is proportional to the product of the number of people infected and the number of people uninfected. If N is the number infected at time t and P is the total number of people in the population, form a differential equation to summarise the scientist's theory.

- b. Separable variables solutions

A differential equation which can be rearranged to the form

$g(y)\frac{dy}{dx} = f(x)$ is said to have **separable variables**.

Example 5

Solve $\frac{dy}{dx} = 2y$.

This DE is similar to the one in Example 1 except here the constant of proportionality is positive. The solution here is one of exponential growth, whereas the needle diagram in Example 1 showed the solution was in the form of exponential decay.

$$\frac{dy}{dx} = 2y$$

$$\frac{1}{y} \frac{dy}{dx} = 2$$

← Separate the variables first.

$$\int \frac{1}{y} dy = \int 2 dx$$

← Show what you are intending to do.

$$\ln y = 2x + c$$

← Integrate both sides.

$$y = Ae^{2x}$$

← Write the solution in its simplest form.

The constant which is added in integration will become a multiplicative constant if there is a log from the integration and you take exponentials (i.e. $A = e^c$) – remember to change the letter when you do this.

Example 6

Solve $\frac{dy}{dx} = 2xy$.

$$\frac{dy}{dx} = 2xy$$

$$\frac{1}{y} \frac{dy}{dx} = 2x$$

$$\int \frac{1}{y} dy = \int 2x dx$$

$$\ln y = x^2 + c$$

$$y = Ae^{x^2}$$

This DE is very similar to that in Example 5. Once the variables are separated you essentially always have two integrations to do and it can require any of the integration techniques you learned in P1 or in Chapters 5 and 8 of this book.

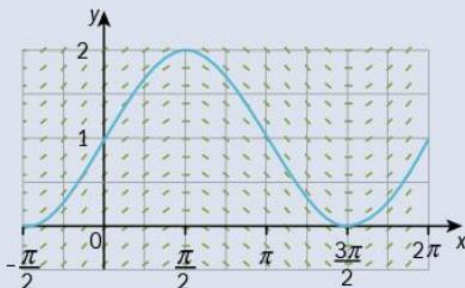
Example 7

Solve $\frac{dy}{dx} = \cos x$.

$$\frac{dy}{dx} = \cos x$$

$$\int 1 dy = \int \cos x dx$$

$$y = \sin x + c$$



The needle diagram shows the gradient function, and the particular solution with $c = 1$ is shown in blue.

Example 8

Solve $\frac{dy}{dx} = \cos^2 y$.

$$\frac{dy}{dx} = \cos^2 y$$

$$\sec^2 y \frac{dy}{dx} = 1$$

$$\int \sec^2 y dy = \int 1 dx$$

$$\tan y = x + c$$

$$y = \tan^{-1}(x + c)$$

← You need to get integrable functions on both sides.

← Integrate both sides.

If the functional form is very complicated then it is not a requirement to give the function of y explicitly, unless the question specifies this. Here $\tan y = x + c$ is a perfectly acceptable form of the solution to the DE.

Example 9

Solve $\frac{dy}{dx} = \frac{3x^2 \sin^2 y}{(x^3 + 2)}$.

$$\frac{dy}{dx} = \frac{3x^2 \sin^2 y}{(x^3 + 2)}$$

$$\operatorname{cosec}^2 y \frac{dy}{dx} = \frac{3x^2}{(x^3 + 2)}$$

$$\int \operatorname{cosec}^2 y \, dy = \int \frac{3x^2}{(x^3 + 2)} \, dx$$

$$\cot y = -\ln A(x^3 + 2), \text{ where } c = \ln A$$

You need to get integrable functions on both sides.

Integrate both sides – you need to remember standard forms of integrable functions.

Example 10

Solve $\frac{dy}{dx} = \frac{2}{x^2 - 1}$.

$$\frac{dy}{dx} = \frac{2}{x^2 - 1}$$

$$\frac{dy}{dx} = \frac{1}{x-1} - \frac{1}{x+1}$$

$$\int dy = \int \left(\frac{1}{x-1} - \frac{1}{x+1} \right) dx$$

$$y = \ln(x-1) - \ln(x+1) + c$$

$$y = \ln \left\{ \frac{A(x-1)}{(x+1)} \right\}$$

You need to use partial fractions to get this into an integrable form.

Here we express c as $\ln A$ in order to combine the terms.

In examinations, you will normally be asked to first express a rational function in partial fractions, and then integrate the partial fractions to solve the DE.

Example 11

Solve $\frac{dy}{dx} = y^2 \ln x$.

$$\frac{dy}{dx} = y^2 \ln x$$

$$\frac{1}{y^2} \frac{dy}{dx} = \ln x$$

$$\int \frac{1}{y^2} \, dy = \int \ln x \, dx$$

$$-\frac{1}{y} = x \ln x - \int x \frac{1}{x} \, dx = x \ln x - x + c$$

Separate the variables.

You will need to integrate this by parts.

Remember, you may be asked to use any of the integration techniques you met in Chapter 8 in solving DEs. After separating the variables you always have two integrations to perform.

c. Finding particular solutions

Example 12

Find the curve which satisfies $\frac{dy}{dx} = 2y$ and passes through $(0, 3)$.

$$\frac{dy}{dx} = 2y$$

$$\Rightarrow y = Ae^{2x}$$

$$x = 0, y = 3 \Rightarrow A = 3$$

$$\Rightarrow y = 3e^{2x}$$

From Example 5: this is the general solution.

Use the initial condition to find A .

This is the particular solution.

Example 13

Find the curve which satisfies $\frac{dy}{dx} = \frac{2}{x^2 - 1}$ and passes through $(2, 0)$.

$$\frac{dy}{dx} = \frac{2}{x^2 - 1}$$

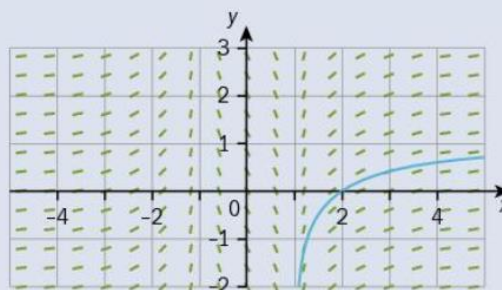
$$y = \ln \left\{ \frac{A(x-1)}{(x+1)} \right\}$$

$$x = 2, y = 0 \Rightarrow 0 = \ln \left(\frac{A}{3} \right) \Rightarrow A = 3$$

$$y = \ln \left\{ \frac{3(x-1)}{(x+1)} \right\}$$

The needle diagram shows the DE with the particular solution through $(2, 0)$.

From Example 10



Example 14

Find the curve which satisfies $\frac{dy}{dx} = \frac{4xy}{(x^2 + 3)}$ and passes through $(0, 9)$.

$$\frac{dy}{dx} = \frac{4xy}{(x^2 + 3)}$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{4x}{(x^2 + 3)}$$

$$\Rightarrow \int \frac{1}{y} dy = \int \frac{4x}{(x^2 + 3)} dx$$

$$\Rightarrow \ln y = 2 \ln(x^2 + 3) + c = \ln A(x^2 + 3)^2$$

$$\Rightarrow y = A(x^2 + 3)^2$$

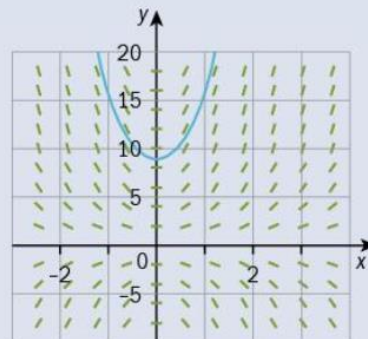
$$\Rightarrow x = 0, y = 9 \Rightarrow A = 1$$

$$\Rightarrow y = (x^2 + 3)^2$$

Separate the variables.

Integrate both sides.

Use the initial condition to find A .



Example 17

Find the curve which satisfies $y^2 \frac{dy}{dx} = x^2 e^{x^3}$ and passes through the origin.

$$y^2 \frac{dy}{dx} = x^2 e^{x^3}$$

← The variables are already separated here.

$$\Rightarrow \int y^2 dy = \int x^2 e^{x^3} dx$$

$$\Rightarrow \frac{1}{3} y^3 = \frac{1}{3} e^{x^3} + c$$

← Since $\frac{d}{dx}(e^{x^3}) = 3x^2 e^{x^3}$

$$x = 0, y = 0 \Rightarrow c = -\frac{1}{3}$$

← Use the initial condition to find the particular solution.

$$\Rightarrow y^3 = e^{x^3} - 1$$

d. Modeling with differential equations

1. A rod has the property that the temperature gradient, $\frac{dT}{dx}$, at a distance x cm from the end of the rod being heated is proportional to the distance x .
The end of the rod is heated to a temperature of 400°C , and when $x = 50$ we know that $\frac{dT}{dx} = -8$.
 - a) Write down a differential equation which is satisfied by the temperature T at a distance x from the end of the rod.
 - b) Solve the differential equation to find an expression for the temperature T in terms of x .
 - c) Calculate the temperature at the point 50 cm from the heated end of the rod.
 - d) Calculate how far from the heated end of the rod the temperature reaches 20°C .
2. In a chemical reaction a substance X reacts with another substance Y .
The masses of substances X and Y after time t seconds are x and y grams respectively.
It is given that $\frac{dy}{dt} = -0.1xy$ and $x = 30e^{-3t}$. Also, when $t = 0$, $y = 20$.
 - a) Form a differential equation in y and t .
 - b) Solve this differential equation to obtain an expression for y in terms of t .
 - c) Find the proportion of Y which will remain when t is very large.